Healthcare Management: Patient Readmission Analysis

End-to-End Data Analysis Project Report

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1. Introduction

Hospital readmissions are a major challenge for healthcare systems worldwide. Frequent readmissions can indicate poor post-discharge care, ineffective treatment plans, or the presence of complex chronic conditions. Reducing unnecessary readmissions helps hospitals improve patient outcomes while also lowering healthcare costs.

This project focuses on analyzing patient data to identify patterns related to **hospital readmissions within 30 days**. Using data analysis tools such as **Excel, SQL, Python, and Tableau**, the project explores patient demographics, treatment complexity, and hospital stay characteristics to identify factors influencing readmission rates.

The goal is to provide actionable insights that can help healthcare providers identify **high-risk patients** and improve care management strategies.

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2. Dataset Overview

The dataset contains healthcare records for diabetic patients, where each row represents a **patient encounter with a healthcare facility**.

Dataset Characteristics

Feature	Description
Total Records	101,766 patient encounters
Total Columns	50 healthcare variables
Data Type	Mixed (numerical and categorical)
Key Focus	Patient readmission within 30 days

Readmission Categories

Category	Meaning
NO	Patient not readmitted

>30	Patient readmitted after 30 days
<30	Patient readmitted within 30 days

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3. Tools and Technologies Used

This project demonstrates an **end-to-end data analytics workflow** using multiple tools.

Tool	Purpose
Excel	Data exploration and summary statistics
SQL (MySQL)	Data querying and aggregation
Python (Pandas, Seaborn, Matplotlib)	Exploratory data analysis and visualization
Tableau	Interactive dashboard and visualization

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4. Excel Analysis

Excel was used for initial **data exploration and descriptive analysis**.

Key Activities

• Generated statistical summaries of numerical features
• Analyzed readmission patterns by gender and age group
• Visualized race distribution
• Created charts for readmission category counts

Key Findings

- The average hospital stay was approximately **4.4 days**.
- Patients received an average of **16 medications per hospital visit**.
- The majority of patients belonged to the **Caucasian ethnic group**.
- Most patients were **not readmitted**, but a significant portion experienced readmissions after discharge.

These insights provided an initial understanding of the dataset before deeper analysis.

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5. SQL Analysis

SQL was used to perform structured queries and extract insights from the dataset.

Key SQL Queries Performed

1. Total number of patient encounters
2. Top 10 most frequent diagnoses
3. Average hospital stay by admission type
4. Percentage of readmitted patients
5. Age distribution of patients
6. Most common procedures
7. Average medications by age group
8. Readmission distribution by payer code

Key Findings

• The dataset contains **101,766 patient encounters**.

• Approximately **46,902 patients were readmitted**, representing a significant proportion of total encounters.

• Older age groups (**60–80 years**) account for the largest number of hospital visits.

• Many encounters involve **multiple diagnoses**, indicating complex patient health conditions.

• The majority of hospital visits involved **no procedures**, suggesting that many cases focused on monitoring, diagnosis, and medication management.

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6. Python Exploratory Data Analysis (EDA)

Python was used to perform deeper exploratory analysis and visualize patterns in the dataset.

Key Analysis Tasks

- Descriptive statistical analysis of numerical variables
- Distribution analysis of categorical variables
- Correlation analysis between numerical features
- Medication change analysis
- Diagnosis distribution analysis
- Admission and discharge patterns
- Outlier detection

Important Insights

Age and Readmission

Older patients showed **higher readmission rates**, especially those between **60–80 years old**.

Medication Patterns

Patients with medication changes often had **higher numbers of medications**, indicating more complex treatment plans.

Diagnosis Patterns

Certain diagnosis codes related to **chronic diseases and cardiovascular conditions** appeared frequently, suggesting strong comorbidity among diabetic patients.

Correlation Analysis

Most numerical features showed **weak to moderate correlations**, suggesting that multiple independent factors contribute to patient readmission.

Outlier Detection

Several extreme values were observed in variables such as:

- number of medications
- lab procedures
- outpatient visits

These outliers likely represent **patients with severe or complex medical conditions**.

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7. Tableau Dashboard

An interactive Tableau dashboard was created to visually explore key factors influencing patient readmission.

Dashboard Components

The dashboard includes:

- **Key Performance Indicators (KPIs)**
- **Interactive filters**
- **Multiple visualizations**

KPIs Included

KPI	Value
Total Patient Encounters	101,766
Total Readmitted Patients	46,902
Average Hospital Stay	4.39 days

KPI	Value
Average Medications per Patient	16.02

Dashboard Visualizations

1. **Readmission vs Diagnoses & Emergency Cases**
   Shows how the number of diagnoses and emergency visits relate to readmission status.
2. **Readmission by Age Group**
   Identifies which age groups experience higher readmission rates.
3. **Healthcare Factors Affecting Readmission**
   Compares medications, procedures, diagnoses, emergency visits, and hospital stay across readmission categories.
4. **Treatment Complexity Bubble Chart**
   Visualizes the relationship between medications, lab procedures, diagnoses, and readmission.

Dashboard Filters

Interactive filters allow users to analyze data by:

- Gender
- Age group
- Readmission status

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8. Key Insights from the Analysis

Several important patterns were identified during the analysis.

1. Age is a major factor in readmissions

Older patients, particularly those aged **60–80 years**, show higher readmission rates.

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2. Treatment complexity increases readmission risk

Patients with:

- more diagnoses
- more medications
- more procedures

tend to have higher readmission counts.

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3. Chronic conditions play a significant role

The most common diagnoses are related to **diabetes complications and cardiovascular diseases**, indicating the impact of chronic illnesses on hospital readmissions.

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4. Emergency visits influence readmission

Patients with frequent emergency visits are more likely to be readmitted.

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5. Hospital stay duration is moderately associated with readmission

Patients with longer hospital stays often have more complex medical conditions, which may increase the likelihood of readmission.

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9. Conclusion

This project demonstrated an end-to-end healthcare data analysis workflow using Excel, SQL, Python, and Tableau.

The analysis identified several key factors influencing patient readmissions, including age, treatment complexity, and chronic health conditions. The interactive Tableau dashboard provides a comprehensive view of these patterns, allowing healthcare providers to explore patient characteristics and identify high-risk groups.

By leveraging data-driven insights, healthcare organizations can improve patient care strategies, reduce unnecessary hospital readmissions, and optimize resource allocation.